

```
1 function [echogram120,fEcho,RangeBinned,pulse_length] =✓
calcEchogram120kHz(Config, Data, WaterProp, Options)
2 % function [echogram120] = calcEchogram120kHz(Config, Data)
3
4 %% INIT
5 echogram120 = [];
6
7 try
8     soundVel = WaterProp.soundVel;
9 catch
10     soundVel = 1500;
11 end
12
13 if isempty(Options)
14 else
15 end
16
17 soundVel2 = soundVel/2;
18
19 %% CALC
20 fAdjMonochromeAmp = 6.0206; % Adjustment used in firmware [dB]
21
22 BD=Config.echo_blanking;
23
24 if (Config.echo_frequency1 == 120)
25     nEcho=1;
26 else
27     if isfield(Config,'echo_frequency2') && Config.echo_frequency2 ==✓
120
28         nEcho=2;
29     else
30         if isfield(Config,'echo_frequency3') && Config.echo_frequency3✓
== 120
31             nEcho=3;
32         else
33             disp('No 120 kHz data');
34             return;
35         end
36     end
37 end
38
```

```
39 % Load filter
40 if Config.fwVersionDoppler >= 2212
41     load CBF120kHz_Tx500us.mat
42     BF120 = CBF_500us;
43 else
44     load CBF120kHz_Tx1ms
45     BF120 = CBF_1ms;    % Firmware versions 2211_4 and older
46 end
47
48
49 L = length(BF120);
50 FiltDelay = round(L/2);
51
52 eval(['sampleRate = Config.echo_rawSampleRate' num2str(nEcho) ';''])
53 eval(['nRawSamp    = Config.echo_nRawSamp'      num2str(nEcho) ';''])
54 eval(['DataI = Data.RawEcho' num2str(nEcho) '_120kHz_DataI;'])
55 eval(['DataQ = Data.RawEcho' num2str(nEcho) '_120kHz_DataQ;'])
56 eval(['Range = Data.RawEcho' num2str(nEcho) '_120kHz_Range;'])
57 eval(['fEcho = Data.Echo' num2str(nEcho) '120_0kHz_Frequency(1);'])
58 eval(['pulse_length = Config.echo_transmitLength' num2str(nEcho) ✓
';'])
59
60 if nRawSamp > 0,
61     sampDist120kHz    = soundVel/(2*sampleRate);
62     Nsamp = size(DataI,1);
63
64     % DSP processing
65     startInd120 = 1+round((BD-Range(1))/sampDist120kHz);
66 else
67     disp('No raw echo data');
68     return;
69 end
70
71 % Apply complex filter in order to reduce noise since the 120 kHz ✓
data
72 % is sampled at 125 kHz with a higher bandwidth than the actual ✓
signal
73 % bandwidth
74 X = DataI(:,startInd120:end)' +j*DataQ(:,startInd120:end)';
75 DataFilt = filter(BF120, 1, X)';
76
```

```

77 sampDist120 = soundVel2/sampleRate;
78 samplesPerCell = Config.echo_binSize/sampDist120;
79 BinLength=zeros(1,Config.echo_numBins);
80
81 % Calculate the number of samples per bin
82 BinLength(1) = round(samplesPerCell);
83 for i=2:Config.echo_numBins
84     BinLength(i) = round((i*Config.echo_binSize - sum(BinLength(1:i-
1))*sampDist120)/sampDist120);
85 end
86
87 echogram120 = zeros(Nsamp, Config.echo_numBins);
88 RangeBinned = zeros(1, Config.echo_numBins);
89 for n=1:Nsamp
90     k = FiltDelay; % Sample number for start of first echogram bin.✓
Blanking has already been accounted for
91         % so start echogram after the filter delay.
92
93     for m=1:Config.echo_numBins
94         samplesInCell = k:(k+BinLength(m)-1);
95
96         Z = DataFilt(n,samplesInCell);
97         sumZZ = sum(Z.*conj(Z));
98         if sumZZ == 0,
99             % Get rid of log10 of zero messages
100             sumZZ = 1e-16;
101         end
102         echogram120(n,m) = fAdjMonochromeAmp + 10*log10✓
(sumZZ/BinLength(m)); % Average. Using 10*log10 since Z is squared
103
104         RangeBinned(m) = mean(Range(startInd120 - FiltDelay +✓
samplesInCell));
105
106         k = k+BinLength(m);
107     end
108 end
109
110 zz=0;
111

```